# Brett H. Andrews

*Curriculum Vitae*

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## Experience

**Research Associate Professor** 2022–present
Department of Physics and Astronomy, University of Pittsburgh

**Research Assistant Professor** 2017–2022
Department of Physics and Astronomy, University of Pittsburgh

**Postdoctoral Associate** 2014–2017
Department of Physics and Astronomy, University of Pittsburgh

## Education

**Ph.D. Astronomy** 2014
The Ohio State University
"Decoding Galaxy Evolution with Gas-phase and Stellar Elemental Abundances"
Advisors: David H. Weinberg & Jennifer A. Johnson

**M.S. Astronomy** 2011
The Ohio State University

**B.S. Physics & Astronomy** 2008
Yale University
Nicolas Adamo Scholar-Athlete Prize (Silliman College)

## Publications

4 lead author; 16 significant contributing author; 63 contributing author; 3 additional publications. Over 18,000 citations. List attached.

## Presentations

5 invited; 32 contributed; 2 posters. List attached.

## Funding

**PI: LSST Discovery Alliance Early Science with Rubin's LSST Fast Turnaround Science Grants** 2026–present
Maximizing Early LSST Science With Deep Learning Photometric Redshifts
**$49,132**

**Co-I: NASA Nancy Grace Roman Space Telescope Research and Support Participation Opportunities** 2025–present
Exploiting the Intrinsic Correlations Between Galaxy Properties: Redshift and Beyond
**$299,887**

**PI: NASA Nancy Grace Roman Space Telescope Research and Support Participation Opportunities** 2023–present
Exploiting Deep Learning to Improve Roman Photometric Redshifts
**$287,881**

**Co-I: NASA Nancy Grace Roman Space Telescope Research and Support Participation Opportunities** 2023–present
A Statistical Framework for Optimizing Roman Spectroscopic Training Sets
**$291,061**

**Co-I: NASA Nancy Grace Roman Space Telescope Research and Support Participation Opportunities** 2023–present
Maximizing Cosmological Science with the Roman High Latitude Imaging Survey
**$19.5M ($714,560 to Pitt)**

**Co-I: NASA Hubble Space Telescope Cycle 30 SNAP Proposal (409 orbits)** 2023–2025
Post-starbursts from DESI: Timing quenching and morphological transformation at $1 < z < 1.3$
**$202,893**

**Senior Personnel: NSF Major Research Instrumentation Program** 2021–2024
Acquisition of Cutting-Edge GPU and MPI Nodes for the Interdisciplinary Pitt Center for Research Computing
**$1,187,606**

**PI: NSF Astronomy and Astrophysics Research Grants** 2020–2025
Interpretable and Deblended Photometric Redshifts with a Deep Capsule Network
**$535,578**

**Co-I: NASA Astrophysical Data Analysis Program** 2019–2024
Multiwavelength Milky Way Analogs
**$504,949**

**Co-I: NASA Astrophysical Data Analysis Program (Funded Extension)** 2021
Multiwavelength Milky Way Analogs
**$33,707**

**NASA PA Space Grant Consortium Research Scholarship Awards** 2019–present
In support of undergraduate research:
Zach Lewis: Fall 2019, Fall 2020, Spring 2021, Fall 2021, Spring 2022
Katie Mack: Summer 2021, Fall 2021, Spring 2022
Emma Moran: Spring 2024, Summer 2024, Fall 2024, Spring 2025
**Total: $32,500**

## Mentoring Ph.D.

**Marcos Tamargo-Arizmendi** (University of Pittsburgh) 2025–present
"Modeling Incompleteness in Spectroscopic Redshift Datasets"

**Finian Ashmead** (University of Pittsburgh) 2023–present
"A Statistical Framework for Optimizing Roman Spectroscopic Training Sets for Roman Photometric Redshifts"

**Yoquelbin Salcedo Hernandez** (University of Pittsburgh) 2022–present
"DESI-2 Emission Line Galaxy Target Selection"

**Ashod Khederlarian** (University of Pittsburgh) 2022–present
"Towards Leveraging Deep Learning for Photometric Redshifts with the Roman Space Telescope"

**Biprateep Dey** (University of Pittsburgh) 2018–2024
"Cosmic Cartography: Photometric Redshifts for Next-Generation Sky Surveys"
*Next Position*: Schmidt AI for Science Fellowship at University of Toronto + CITA Fellowship + Dunlap Institute Fellowship

**Troy Raen** (University of Pittsburgh) 2019–2022
"Toward the Study of Stars with LSST"
*Next Position*: Application Developer at IPAC/Caltech

**Catherine Fielder** (University of Pittsburgh) 2018–2022
"Constraining the Milky Way's Ultraviolet to Infrared SED"
*Next Position*: Postdoctoral Researcher at University of Arizona

**Ph.D. Thesis Committees** (University of Pittsburgh)
– Nathalie Chicoine 2026–present
– Julissa Sarmiento 2025–present

M.S.

**Quanbin (Eric) Ma** (Carnegie Mellon University Machine Learning Department) 2017
"Deep Dimensionality Reduction of MaNGA Data"
*Next Position*: Software Engineer at Facebook

Undergraduate

**Emma Moran** (University of Pittsburgh) 2023–2025
"Deep Learning Improves Photometric Redshifts in All Regions of Color Space"
*Next Position*: NSF GRFP Fellow and Astronomy Ph.D. Student at the Ohio State University

**Zach Lewis** (University of Pittsburgh) 2019–2022
"The Gas-Phase Mass-Metallicity Relation for Massive Galaxies at $z \sim 0.7$ with the LEGA-C Survey"
*Next Position*: NSF GRFP Fellow and Astronomy Ph.D. Student at the University of Wisconsin

**Katie Mack** (University of Pittsburgh) 2021–2022
"Comparing Stellar and Emission Line Dust Attenuation in the LEGA-C Survey"

**Ian Cooper** (University of Pittsburgh) 2016
"SNIa Hosts in MaNGA"
*Next Position*: Geophysics Ph.D. Student at Boston College

High School

**Mariah Jones** (Baldwin High School) 2020–2021
"The Mass–Metallicity Relation at Intermediate Redshifts"
*Next Position*: Undergraduate Student at Vassar College

## Service to Profession

**Invited Reviewer**

– *NSF*
– *NASA*
– *Swiss NSF*

**Invited Referee**
– *The Astrophysical Journal*
– *Monthly Notices of the Royal Astronomical Society*
– *Astronomy & Astrophysics*
– *Publications of the Astronomical Society of the Pacific*

**Faculty Hiring Committee**
Department of Physics and Astronomy, University of Pittsburgh
– LSST LINCC Research Assistant Professor 2022–2023
2021–2022
– Astrophysics or Experimental High Energy Physics with Machine Learning (Tenure-stream) 2018–2019

**Research Faculty Promotion Criteria Committee** 2024
University of Pittsburgh Dietrich School of Arts and Sciences

**Union of Pitt Faculty (USW Local 1088-04)**
University of Pittsburgh
– Research Funding Caucus (Co-chair) 2026–present
– Research Stream Advocacy Committee (Founding Chair) 2025–present
– Bargaining Committee 2025–present
– Organizing Committee 2024–present
– Steward 2024–present

**Faculty Peer Buddy** 2021–2023
University of Pittsburgh

**Faculty Mentor** 2025
LSST Dark Energy Science Collaboration (DESC)

**Astrophysics Academic Program Committee** 2025
University of Pittsburgh

## Teaching

**ASTRON 3580: Galactic and Extragalactic Astronomy** (University of Pittsburgh) Spring 2024
Co-instructor with Jeff Newman.

**Member of the Graduate Faculty** (University of Pittsburgh) 2024–present

## Workshops

**AstroPGH Python Boot Camp and Summer Seminar Series** (University of Pittsburgh and Carnegie Mellon University)
• Intense three day workshop introducing the Python programming language for astrophysics/physics graduate and undergraduate students.
– Organizer 2025
– Organizer. Presented lecture on Matplotlib. 2024
– Organizer 2023
– Organizer. Presented lecture on Python and Jupyter. 2022
– Organizer. Presented two lectures on Numpy and Pandas. 2021
– Organizer. Presented eight lectures on Python basics, Astropy, Pandas, Git, and GitHub. 2020

**Writing Fridays Workshops** (University of Pittsburgh) 2023–present
• Led weekly writing workshops for astrophysics graduate students.

**Professional Development Workshops** (University of Pittsburgh)
• Co-led workshops with Rachel Bezanson for astrophysics and physics graduate students.
– CV, Resume, and Webpage Workshop 2024
– Webpage and CV Workshops 2022

– Webpage Workshop 2021
– CV, Resume, and Webpage Workshops 2020

**Data Science Group Meeting** (University of Pittsburgh) 2018
• Organized event series for astrophysics graduate students.

**Marvin Workshops** (SDSS-IV Collaboration Meetings) 2015–2017
• Led introductory workshops for users of the `Marvin` software toolkit.

**Python Boot Camps** (University of Pittsburgh) 2015
• Led events for undergraduate and graduate students.

## Science Communication

**The Ellis School**

– Third Grade STEM Class: "The Nancy Grace Roman Space Telescope" 2026
– First Grade STEM Class: "The Nancy Grace Roman Space Telescope" 2026
– First Grade STEM Class: "How the Sun, Earth, and Moon align for a solar eclipse" 2024

**Shadyside Presbyterian Church Nursery School**
– Pre-K Class: "A Few Beautiful Minutes: Experiencing a Solar Eclipse" 2024

**Allegheny Observatory Public Lecture Series**
– "The True Colors of the Milky Way" 2023

**Astronomy on Tap**
– "The Origin of the Elements" 2018
– "Indiana Jones and the Hidden Galaxy" w/ Courtney Epstein 2014

**Learn & Earn Corporate Host** 2021
Hosted intern for summer youth employment program delivered by Allegheny County, the City of Pittsburgh, and Partner4Work.

**Science Olympiad** 2011
Coached Grandview Heights (Ohio) Middle School team.

## Collab.

**Roman Science Collaboration Member** 2025–present

**Roman High Latitude Imaging Survey (HLIS) Cosmology Project Infrastructure Team (PIT) Member** 2024–present
– CAS WG co-chair 2026–present

**Dark Energy Spectroscopic Instrument (DESI) Member** 2024–present

**LSST Dark Energy Science Collaboration (DESC) Full Member** 2023–present

**SDSS-IV/MaNGA Architect** 2016

## Software

**`flexCE`**
Python package for modeling galactic chemical evolution.

**`Marvin`**
Python package, RESTful API, and Flask web application for accessing, visualizing, and analyzing SDSS-IV MaNGA integral-field spectroscopic data.

# Brett H. Andrews

*Publications*

**Lead Author**

4. **Andrews, B. H.**, et al., 2017, *The Astrophysical Journal*, 835, 224
   Inflow, Outflow, Yields, and Stellar Population Mixing in Chemical Evolution Models
3. **Andrews, B. H.** & Martini, P., 2013, *The Astrophysical Journal*, 765, 140
   The Mass-Metallicity Relation with the Direct Method on Stacked Spectra of SDSS Galaxies
2. **Andrews, B. H.**, et al., 2012, *Acta Astronomica*, 62, 269
   Principal Component Abundance Analysis of Microlensed Bulge Dwarf and Subgiant Stars
1. **Andrews, B. H.** & Thompson, T. A., 2011, *The Astrophysical Journal*, 727, 97
   Assessing Radiation Pressure as a Feedback Mechanism in Star-forming Galaxies

**Sig. Contrib. Author**

16. Dey, B., et al., 2026, arXiv:2604.06143, submitted to *The Astronomical Journal*
    Deep Spectroscopy with DESI for Photometric Redshift Training and Calibration
15. Salcedo Hernandez, Y., Newman, J. A., **Andrews, B. H.**, et al., 2026, *The Astronomical Journal*, 172, 11
    Highly Efficient Selection of High-Redshift Emission-Line Galaxies for future DESI-like surveys with Deep Multi-band Imaging
14. Khederlarian, A., **Andrews, B. H.**, Newman, J. A., et al., 2026, *The Astronomical Journal*, 171, 340
    Optimizing Deep Learning Photometric Redshifts for the Roman Space Telescope with HST/CANDELS
13. Moran, E. R., **Andrews, B. H.**, Newman, J. A., et al., 2026, *The Astrophysical Journal*, 997, 36
    Deep Learning Improves Photometric Redshifts in All Regions of Color Space
12. Ashmead, F., Newman, J. A., **Andrews, B. H.**, et al., 2025, arXiv:2512.09032, submitted to *The Astrophysical Journal*
    Optimizing Photometric Redshift Training Sets I: Efficient Compression of the Galaxy Color-Redshift Relation with UMAP
11. Dey, B., Zhao, D., **Andrews, B. H.**, et al., 2025, *Machine Learning: Science and Technology*, 6, 045058
    Towards instance-wise calibration: local amortized diagnostics and reshaping of conditional densities (LADaR)
10. Khederlarian, A., Newman, J. A., **Andrews, B. H.**, et al., 2024, *Monthly Notices of the Royal Astronomical Society*, 531, 1454
    Emission Line Predictions for Mock Galaxy Catalogues: a New Differentiable and Empirical Mapping from DESI
9. Lewis, Z., **Andrews, B. H.**, et al., 2024, *The Astrophysical Journal*, 964, 59
   The Gas-Phase Mass–Metallicity Relation for Massive Galaxies at $z \sim 0.7$ with the LEGA-C Survey
8. Fielder, C., **Andrews, B. H.**, et al., 2023, *Monthly Notices of the Royal Astronomical Society*, 525, 1023
   Empirically Driven Multiwavelength K-corrections At Low Redshift
7. Kodra, D., **Andrews, B. H.**, et al., 2023, *The Astrophysical Journal*, 942, 36
   Optimized Photometric Redshifts for the Cosmic Assembly Near-Infrared Deep Extragalactic Legacy Survey (CANDELS)
6. Dey, B., **Andrews, B. H.**, et al., 2022, *Monthly Notices of the Royal Astronomical Society*, 515, 4
   Photometric Redshifts from SDSS Images with an Interpretable Deep Capsule Network
5. Dey, B., Newman, J. A., **Andrews, B. H.**, et al., 2021, *arXiv:2110.15209*
   Re-calibrating Photometric Redshift Probability Distributions Using Feature-space Regression

4. Fielder, C. E., Newman, J. A., **Andrews, B. H.**, et al., 2021, *Monthly Notices of the Royal Astronomical Society*, 508, 4459
Constraining the Milky Way's ultraviolet-to-infrared SED with Gaussian process regression
3. Cherinka, B., **Andrews, B. H.**, et al., 2019, *The Astronomical Journal*, 158, 74
Marvin: A Tool Kit for Streamlined Access and Visualization of the SDSS-IV MaNGA Data Set
2. Weinberg, D. H., **Andrews, B. H.**, & Freudenburg, J., 2017, *The Astrophysical Journal*, 837, 183
Equilibrium and Sudden Events in Chemical Evolution
1. Brown, J. S., Martini, P., & **Andrews, B. H.**, 2016, *Monthly Notices of the Royal Astronomical Society*, 458, 1529
A recalibration of strong-line oxygen abundance diagnostics via the direct method and implications for the high-redshift universe

Contributing Author

63. Zhang, T., et al., 2025, arXiv:2510.07370, accepted to *Monthly Notices of the Royal Astronomical Society*
Photometric Redshift Estimation for Rubin Observatory Data Preview 1 with Redshift Assessment Infrastructure Layers (RAIL)
62. de Souza, D., et al., 2026, arXiv:2602.09230, submitted to *Journal of Cosmology and Astroparticle Physics*
Modeling Redshift Uncertainties in Roman Weak Lensing Cosmology
61. Scholte, D., et al., 2026, arXiv:2601.02463, submitted to *Monthly Notices of the Royal Astronomical Society*
Electron temperature relations and the direct N, O, Ne, S and Ar abundances of 49959 star-forming galaxies in DESI Data Release 2
60. Zhang, T., et al., 2026, *Monthly Notices of the Royal Astronomical Society*, 545, 1829
Forecasting the Impact of Source Galaxy Photometric Redshift Uncertainties on the LSST $3 \times 2$pt Analysis
59. Zhang, Y., et al., 2024, *The Astrophysical Journal*, 976, 36
DESI Massive Post-Starburst Galaxies at $z \sim 1.2$ have compact structures and dense cores
58. Moskowitz, I., et al., 2024, *The Astrophysical Journal Letters*, 967, L6
Improving Photometric Redshift Estimates with Training Sample Augmentation
57. Yantovski-Barth, M. J., et al., 2024, *Monthly Notices of the Royal Astronomical Society*, 531, 2285
The CluMPR Galaxy Cluster-Finding Algorithm and DESI Legacy Survey Galaxy Cluster Catalogue
56. Zhou, S., et al., 2023, *Monthly Notices of the Royal Astronomical Society*, 521, 5810
Are Milky-Way-like galaxies like the Milky Way? A view from SDSS-IV/MaNGA
55. Setton, D., et al., 2023, *The Astrophysical Journal Letters*, 947, 31
DESI Survey Validation Spectra Reveal an Increasing Fraction of Recently Quenched Galaxies at $z \sim 1$
54. Kartaltepe, J., et al., 2023, *The Astrophysical Journal Letters*, 946, 15
CEERS Key Paper. III. The Diversity of Galaxy Structure and Morphology at z = 3–9 with JWST
53. Boardman, N., et al., 2022, *Monthly Notices of the Royal Astronomical Society*, 514, 2298
How well do local relations predict gas-phase metallicity gradients? Results from SDSS-IV MaNGA
52. Oyarzún, G. A., et al., 2022, *The Astrophysical Journal*, 933, 88
SDSS-IV MaNGA: How the Stellar Populations of Passive Central Galaxies Depend on Stellar and Halo Mass
51. Schaefer, A. L., et al., 2022, *The Astrophysical Journal*, 930, 2
SDSS-IV MaNGA: Exploring the Local Scaling Relations for N/O

50. Abdurro'uf, et al., 2021, *The Astrophysical Journal Supplement Series*, 259, 2
The Seventeenth Data Release of the Sloan Digital Sky Surveys: Complete Release of MaNGA, MaStar, and APOGEE-2 Data
49. Law, D., et al., 2020, *The Astrophysical Journal*, 915, 35
SDSS-IV MaNGA: Refining Strong Line Diagnostic Classifications Using Spatially Resolved Gas Dynamics
48. Parikh, T., et al., 2021, *Monthly Notices of the Royal Astronomical Society*, 502, 5508
SDSS-IV MaNGA: radial gradients in stellar population properties of early-type and late-type galaxies
47. Greener, M., et al., 2021, *Monthly Notices of the Royal Astronomical Society*, 502, 95
SDSS-IV MaNGA: the 'G-dwarf problem' revisited
46. Luo, Y., et al., 2021, *The Astrophysical Journal*, 908, 183
Evidence for the Accretion of Gas in Star-forming Galaxies: High N/O Abundances in Regions of Anomalously Low Metallicity
45. Mazzola, C., et al., 2020, *Monthly Notices of the Royal Astronomical Society*, 499, 1607
The close binary fraction as a function of stellar parameters in APOGEE: a strong anticorrelation with $\alpha$ abundances
44. Boardman, N., et al., 2020, *Monthly Notices of the Royal Astronomical Society*, 498, 4943
Are the Milky Way and Andromeda unusual? A comparison with Milky Way and Andromeda analogues
43. Fraser-McKelvie, A., et al., 2020, *Monthly Notices of the Royal Astronomical Society*, 495, 4158
SDSS-IV MaNGA: spatially resolved star formation in barred galaxies
42. Ahumada, R., et al., 2020, *The Astrophysical Journal Supplement Series*, 249, 3
The 16th Data Release of the Sloan Digital Sky Surveys: First Release from the APOGEE-2 Southern Survey and Full Release of eBOSS Spectra
41. Greener, M. J., et al., 2020, *Monthly Notices of the Royal Astronomical Society*, 495, 2305
SDSS-IV MaNGA: spatially resolved dust attenuation in spiral galaxies
40. Schaefer, A. L., et al., 2020, *The Astrophysical Journal*, 890, L3
SDSS-IV MaNGA: Variations in the N/O-O/H Relation Bias Metallicity Gradient Measurements
39. Boardman, N., et al., 2020, *Monthly Notices of the Royal Astronomical Society*, 491, 3672
Milky Way analogues in MaNGA: multiparameter homogeneity and comparison to the Milky Way
38. Westfall, K. B., et al., 2019, *The Astronomical Journal*, 158, 231
The Data Analysis Pipeline for the SDSS-IV MaNGA IFU Galaxy Survey: Overview
37. Fraser-McKelvie, A., et al., 2019, *Monthly Notices of the Royal Astronomical Society*, 488, L6
SDSS-IV MaNGA: stellar population gradients within barred galaxies
36. Pace, Z. J., et al., 2019, *The Astrophysical Journal*, 883, 83
Resolved and Integrated Stellar Masses in the SDSS-IV/MaNGA Survey. II. Applications of PCA-based Stellar Mass Estimates
35. Pace, Z. J., et al., 2019, *The Astrophysical Journal*, 883, 82
Resolved and Integrated Stellar Masses in the SDSS-IV/MaNGA Survey. I. PCA Spectral Fitting and Stellar Mass-to-light Ratio Estimates
34. Zhang, K., et al., 2019, *The Astrophysical Journal*, 883, 63
Machine-learning Classifiers for Intermediate Redshift Emission-line Galaxies
33. Oyarzún, G. A., et al., 2019, *The Astrophysical Journal*, 880, 111
Signatures of Stellar Accretion in MaNGA Early-type Galaxies
32. Parikh, T., et al., 2019, *Monthly Notices of the Royal Astronomical Society*, 483, 3420
SDSS-IV MaNGA: local and global chemical abundance patterns in early-type galaxies
31. Aguado, D. S., et al., 2019, *The Astrophysical Journal Supplement Series*, 240, 23
The Fifteenth Data Release of the Sloan Digital Sky Surveys: First Release of MaNGA-derived Quantities, Data Visualization Tools, and Stellar Library
30. Hwang, H.-C., et al., 2019, *The Astrophysical Journal*, 872, 144
Anomalously Low-metallicity Regions in MaNGA Star-forming Galaxies: Accretion Caught in Action?

29. Li, H., et al., 2019, *The Astrophysical Journal*, 872, 63
Interpreting the Star Formation-Extinction Relation with MaNGA
28. Zasowski, G., et al., 2019, *The Astrophysical Journal*, 870, 138
APOGEE DR14/DR15 Abundances in the Inner Milky Way
27. Rowlands, K., et al., 2018, *Monthly Notices of the Royal Astronomical Society*, 480, 2544
SDSS-IV MaNGA: spatially resolved star formation histories and the connection to galaxy physical properties
26. Parikh, T., et al., 2018, *Monthly Notices of the Royal Astronomical Society*, 477, 3954
SDSS-IV MaNGA: the spatially resolved stellar initial mass function in ∼400 early-type galaxies
25. Penny, S. J., et al., 2018, *Monthly Notices of the Royal Astronomical Society*, 476, 979
SDSS-IV MaNGA: evidence of the importance of AGN feedback in low-mass galaxies
24. Abolfathi, B., et al., 2018, *The Astrophysical Journal Supplement Series*, 235, 42
The Fourteenth Data Release of the Sloan Digital Sky Survey: First Spectroscopic Data from the Extended Baryon Oscillation Spectroscopic Survey and from the Second Phase of the Apache Point Observatory Galactic Evolution Experiment
23. Talbot, M. S., et al., 2018, *Monthly Notices of the Royal Astronomical Society*, 477, 195
SDSS-IV MaNGA: the spectroscopic discovery of strongly lensed galaxies
22. Wylezalek, D., et al., 2018, *Monthly Notices of the Royal Astronomical Society*, 474, 1499
SDSS-IV MaNGA: identification of active galactic nuclei in optical integral field unit surveys
21. Badenes, C., et al., 2018, *The Astrophysical Journal*, 854, 147
Stellar Multiplicity Meets Stellar Evolution and Metallicity: The APOGEE View
20. Albareti, F. D., et al., 2017, *The Astrophysical Journal Supplement Series*, 233, 25
The 13th Data Release of the Sloan Digital Sky Survey: First Spectroscopic Data from the SDSS-IV Survey Mapping Nearby Galaxies at Apache Point Observatory
19. Greene, J. E., et al., 2017, *The Astrophysical Journal*, 851, L33
SDSS-IV MaNGA: Probing the Kinematic Morphology-Density Relation of Early-type Galaxies with MaNGA
18. Zasowski, G., et al., 2017, *The Astronomical Journal*, 154, 198
Target Selection for the SDSS-IV APOGEE-2 Survey
17. Zahid, H. J., et al., 2017, *The Astrophysical Journal*, 847, 18
Stellar Absorption Line Analysis of Local Star-forming Galaxies: The Relation between Stellar Mass, Metallicity, Dust Attenuation, and Star Formation Rate
16. Martínez-Rodríguez, H., et al., 2017, *The Astrophysical Journal*, 843, 35
Observational Evidence for High Neutronization in Supernova Remnants: Implications for Type Ia Supernova Progenitors
15. Blanton, M. R., et al., 2017, *The Astronomical Journal*, 154, 28
Sloan Digital Sky Survey IV: Mapping the Milky Way, Nearby Galaxies, and the Distant Universe
14. Linden, S. T., et al., 2017, *The Astrophysical Journal*, 842, 49
Timing the Evolution of the Galactic Disk with NGC 6791: An Open Cluster with Peculiar High-$\alpha$ Chemistry as Seen by APOGEE
13. Zhang, K., et al., 2017, *Monthly Notices of the Royal Astronomical Society*, 466, 3217
SDSS-IV MaNGA: the impact of diffuse ionized gas on emission-line ratios, interpretation of diagnostic diagrams and gas metallicity measurements
12. Jones, A., et al., 2017, *Astronomy and Astrophysics*, 599, A141
SDSS IV MaNGA: Deep observations of extra-planar, diffuse ionized gas around late-type galaxies from stacked IFU spectra
11. Yan, R., et al., 2016, *The Astronomical Journal*, 152, 197
SDSS-IV MaNGA IFS Galaxy Survey—Survey Design, Execution, and Initial Data Quality
10. Law, D. R., et al., 2016, *The Astronomical Journal*, 152, 83
The Data Reduction Pipeline for the SDSS-IV MaNGA IFU Galaxy Survey
9. Holtzman, J. A., et al., 2015, *The Astronomical Journal*, 150, 148
Abundances, Stellar Parameters, and Spectra from the SDSS-III/APOGEE Survey

8. Hayden, M. R., et al., 2015, *The Astrophysical Journal*, 808, 132
   Chemical Cartography with APOGEE: Metallicity Distribution Functions and the Chemical Structure of the Milky Way Disk
7. Alam, S., et al., 2015, *The Astrophysical Journal Supplement Series*, 219, 12
   The Eleventh and Twelfth Data Releases of the Sloan Digital Sky Survey: Final Data from SDSS-III
6. Nidever, D. L., et al., 2014, *The Astrophysical Journal*, 796, 38
   Tracing Chemical Evolution over the Extent of the Milky Way's Disk with APOGEE Red Clump Stars
5. Bovy, J., et al., 2014, *The Astrophysical Journal*, 790, 127
   The APOGEE Red-clump Catalog: Precise Distances, Velocities, and High-resolution Elemental Abundances over a Large Area of the Milky Way's Disk
4. Ahn, C. P., et al., 2014, *The Astrophysical Journal Supplement Series*, 211, 17
   The Tenth Data Release of the Sloan Digital Sky Survey: First Spectroscopic Data from the SDSS-III Apache Point Observatory Galactic Evolution Experiment
3. Leja, J., et al., 2013, *The Astrophysical Journal*, 778, L24
   Exploring the Chemical Link between Local Ellipticals and Their High-redshift Progenitors
2. Zasowski, G., et al., 2013, *The Astronomical Journal*, 146, 81
   Target Selection for the Apache Point Observatory Galactic Evolution Experiment (APOGEE)
1. Ahn, C. P., et al., 2012, *The Astrophysical Journal Supplement Series*, 203, 21
   The Ninth Data Release of the Sloan Digital Sky Survey: First Spectroscopic Data from the SDSS-III Baryon Oscillation Spectroscopic Survey

Additional

3. Cherinka, B., et al., 2020, *ASPC*, 527, 743C
   Marvin: A Toolkit for Streamlined Access and Visualisation of the SDSS-IV MaNGA Data Set
2. Wilson, J. C., et al., 2012, *SPIE*, 8446, 84460H
   Performance of the Apache Point Observatory Galactic Evolution Experiment (APOGEE) high-resolution near-infrared multi-object fiber spectrograph
1. **Andrews, B. H.** & Thompson, T. A., 2011, *EAS Publications Series*, 52, 275
   Radiation Pressure Feedback in Galaxies

# Brett H. Andrews

*Presentations*

## Invited

Optimizing Roman Photometric Redshifts for HLIS — 2025
Roman Virtual Lecture Series

Effective Plotting — 2021
DESI Collaboration Meeting

Hidden in Plain Sight: a Deep Learning Approach to Finding Supernovae in Galaxy Maps — 2018
Science 2018, University of Pittsburgh

The SDSS-IV MaNGA Survey: Galaxy Dissection on an Industrial Scale — 2017
CMU Astro Seminar, Carnegie Mellon University

The Mass–Metallicity Relation in SDSS Using Electron Temperature Measurements — 2015
Understanding Nebular Emission in High-Redshift Galaxies, The Carnegie Observatories

## Contributed

Photometric Redshifts for Roman HLIS — 2026
Roman Science Quarterly, Space Telescope Science Institute

Optimizing Roman Photometric Redshifts for HLIS — 2025
Cosmic Cartography with Roman, Space Telescope Science Institute

Optimizing Roman Photometric Redshifts — 2024
Roman HLIS Cosmology PIT Meeting, California Institute of Technology

Photometric Redshifts for Next Generation Imaging Surveys — 2024
DHWFest, University of Utah

Jumpstart Your Paper — 2022
AstroPGH Summer Seminar, University of Pittsburgh

AstroCoffee Tips — 2022
AstroPGH Summer Seminar, University of Pittsburgh

Leveraging Statistics and Machine Learning for Probing Galaxy Evolution and Measuring Galaxy Distances — 2022
AstroLunch Seminar, University of Pittsburgh

Effective Plotting — 2021
AstroPGH Summer Seminar, University of Pittsburgh

Effective Plotting — 2020
AstroPGH Summer Seminar, University of Pittsburgh

Effective Plotting — 2018
Astro Student Seminar, University of Pittsburgh

GitHub Flow — 2018
Astro Student Seminar, University of Pittsburgh

MaNGA DAP — 2017
SDSS-IV Collaboration Meeting, Pontificia Universidad Católica de Chile

Streamlining MaNGA Data with Marvin — 2017
AstroLunch, University of Pittsburgh

The State of Marvin — 2016
SDSS-IV Collaboration Meeting, University of Wisconsin

AstroCoffee Presentation Advice — 2016
Astro Student Seminar, University of Pittsburgh

Marvin-tools: Distilling DAP Measurements and & Stacking Spectra — 2016
MaNGA Collaboration Meeting (Cocoa Beach, FL)

Global vs. Resolved Metallicities — 2015
Astro Student Seminar, University of Pittsburgh

Global vs. Resolved Metallicities — 2015
SDSS-IV Collaboration Meeting, Instituto Física Teórica Universidad Autónoma de Madrid

MaNGA Data Analysis Pipeline Quality Assessment — 2015
MaNGA Collaboration Meeting, University of Kentucky

Principal Component Abundance Analysis: APOGEE and flexCE — 2015
Local Group Astrostatistics Conference, University of Michigan

Understanding the Bimodality in [alpha/Fe] with a Chemical Evolution Model — 2014
SDSS-III Collaboration Meeting (Park City, UT)

Applying Principal Component Analysis to APOGEE — 2014
SDSS-IV Collaboration Meeting (Park City, UT)

Exploiting Large Multi-element Stellar Abundance Surveys — 2014
AAS Winter Meeting

Decoding Galactic Chemical Evolution with Gas-phase and Stellar Abundances — 2013
Colloquium, University of Wisconsin

Yields, Delays, and Mixing in Chemical Evolution Models — 2013
SDSS-II Collaboration Meeting, Johns Hopkins University

Decoding Galactic Chemical Evolution with Gas-phase and Stellar Abundances — 2013
Yale Center for Astronomy & Astrophysics Seminar, Yale University

The Mass–Metallicity Relation: A Window Into Galaxy Evolution — 2013
Seminar, National Radio Astronomical Observatory

The Galaxy Mass–Metallicity Relation: A Sensitive Diagnostic of the Processes That Drive Galaxy Evolution — 2013
Edward F. Hayes Graduate Research Forum, The Ohio State University

Characterizing the Distribution of Stars in Chemical Abundance Space — 2013
APOGEE Collaboration Meeting, The Carnegie Observatories

Accounting for Stellar Abundance Uncertainties in Principal Component Abundance Analysis — 2012
APOGEE Collaboration Meeting, Texas Christian University

Principal Component Abundance Analysis: Finding the Principal Components of Chemical Abundance Space — 2011
SDSS-III Collaboration Meeting, Vanderbilt University

Radiation Pressure Feedback in Galaxies — 2011
Edward F. Hayes Graduate Research Forum, The Ohio State University

## Posters

Principal Component Abundance Analysis and Chemical Evolution Models — 2014
The Near-Field Deep-Field Connection, University of California, Irvine

Assessing Radiation Pressure as a Feedback Mechanism in Star-forming Galaxies — 2011
The 5th Zermatt ISM Symposium